TECHNICAL REQUIREMENT DOCUMENT (TRD)

Architecture, Mathematical Physics Engine, Machine Learning Infrastructure, and Control Integration

Heavy Oil Wells – Baghewala Field, Rajasthan

Smart India Hackathon (SIH) 2026

System Architecture & Technical Specification

September 5, 2026

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1 System Architecture & Technical Overview

1.1 High-Level Component Architecture

The Well-to-Surface Digital Twin architecture for Baghewala Field is structured as an integrated multi-tier edge-cloud ecosystem. It bridges physical wellhead telemetry with high-performance computational models, physics-informed machine learning (PINN), and real-time Variable Frequency Drive (VFD) actuations.

1. **Edge Layer (Field Gateway & Modbus/MQTT Ingestion)**: Deployed directly at the wellsite via ruggedized Industrial IoT (IIoT) edge hardware (e.g., Siemens MindConnect / Advantech UNO). Runs local micro-services for high-frequency load-position filtering, local dynamic safe speed calculation (SPM_{safe}), and safe autonomous fallback control.
2. **Ingestion & Messaging Backbone**: Apache Kafka distributed event streaming bus paired with EMQX MQTT broker for real-time telemetry streaming (≥ 10 Hz).
3. **Physics Computations & Numerical Engine**: C++/Python-compiled containerized solver running explicit/implicit Finite Difference schemes for 1D Damped Wave Equation wave mechanics (Gibbs model) and numerical heat transfer.
4. **AI/ML Diagnostic & Optimization Pipeline**: Hybrid architecture combining PyTorch-based Physics-Informed Neural Networks (PINNs), ResNet-18 Dyno Card image classifiers, and Reinforcement Learning (PPO) control policies.
5. **Data Persistence Tier**: Dual-database setup utilizing TimescaleDB for continuous high-rate time-series telemetry and PostgreSQL for relational asset metadata, failure logs, and CSS injection records.

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2 Data Pipeline & Data Engineering Architecture

2.1 Field Ingestion Protocols & Telemetry Schema

Telemetry is collected from wellhead SCADA, load cells, motor encoders, and surface pressure-temperature sensors via Modbus RTU/TCP converted to MQTT messages over LTE/Industrial Private 5G.

2.1.1 Telemetry Payload JSON Schema (MQTT Topic: `baghewala/well-07/telemetry`)


```
{
  "well_id": "BGW-JH-07",
  "timestamp": "2026-09-05T19:45:00.120Z",
  "surface_metrics": {
    "polished_rod_position_in": 86.4,
    "polished_rod_load_lbs": 14250.0,
    "motor_current_amp": 42.8,
    "vfd_frequency_hz": 48.2,
    "spm_actual": 5.4,
    "wellhead_temp_c": 84.5,
    "wellhead_pressure_psi": 165.2
  },
  "sensor_health": {
    "load_cell_status": "HEALTHY",
    "position_encoder_status": "HEALTHY"
  }
}
```

2.2 Signal Conditioning & Preprocessing Pipeline

Raw force-displacement data contains mechanical vibrations and electrical noise from the VFD. The ingestion worker executes a real-time digital filtering pipeline before feeding dyno card reconstruction algorithms:

1. **Zero-Phase Butterworth Low-Pass Filter**: 4^{th} order filter with cut-off frequency $f_c = 2.5$ Hz to eliminate non-mechanical structural noise.
2. **Savitzky-Golay Moving Polynomial Smooth**: Window size $N = 11$, polynomial order $p = 2$ applied across load samples to compute smooth dF/dt derivative profiles.
3. **Position Normalization**: Alignment of top-dead-center (TDC) and bottom-dead-center (BDC) positions using zero-crossing detection on velocity $v_{\text{rod}}(t) = dx/dt$.

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3 Computational Engine & Mathematical Physics Implementation

3.1 1D Damped Wave Mechanics Numerical Solver (Gibbs Equation)

To reconstruct the downhole dynamometer card at the pump plunger depth ($x = L$), the system solves the second-order partial differential equation (PDE) governing elastic rod movement:

$$\frac{\partial^2 u(x,t)}{\partial t^2} = a^2 \frac{\partial^2 u(x,t)}{\partial x^2} - c(t)\frac{\partial u(x,t)}{\partial t} \tag{1}$$

where $u(x,t)$ is rod axial displacement, $a = \sqrt{E/\rho_{\text{steel}}} \approx 5000$ m/s is acoustic velocity in steel, and $c(t)$ is viscous fluid damping coefficient:

$$c(t) = \frac{2\pi\mu(t)}{\rho_{\text{steel}} A_{\text{rod}} \ln(D_{\text{tubing}}/D_{\text{rod}})} \tag{2}$$

3.1.1 Discretization via Finite Difference Scheme

The rod string of total length L is discretized into M spatial steps $\Delta x = L/M$ and time step Δt. Using central difference approximations for temporal and spatial second derivatives and backward difference for damping:

$$u_i^{j+1} = \frac{1}{1 + c(t)\Delta t}\left[2u_i^j - u_i^{j-1}(1 - c(t)\Delta t) + \left(\frac{a\Delta t}{\Delta x}\right)^2 \left(u_{i+1}^j - 2u_i^j + u_{i-1}^j\right)\right] \tag{3}$$

Boundary Conditions:

- Top boundary ($x = 0$): $u(0,t) = x_{\text{surface}}(t)$ (measured polished rod position).
- Force top boundary ($x = 0$): $EA_{\text{rod}} \frac{\partial u}{\partial x}\big|_{x=0} = F_{\text{surface}}(t)$ (measured polished rod load).
- Downhole bottom boundary ($x = L$): Downhole load $F_{\text{plunger}}(t) = EA_{\text{rod}} \frac{\partial u}{\partial x}\big|_{x=L}$ and position $u_{\text{plunger}}(t) = u(L,t)$.

3.2 Coupled Thermodynamic & Fluid Viscosity Solver

3.2.1 Thermal Decay Module

Computes near-wellbore spatial temperature $T(t)$ across production elapsed time t (days) post-steam injection:

$$T(t) = T_{\text{res}} + (T_{\text{steam}} - T_{\text{res}})\exp(-\alpha t) \tag{4}$$

where α is dynamically estimated from production thermal mass enthalpy flow:

$$\alpha = \frac{q_o(t) \cdot \rho_o \cdot C_{po} + q_w(t) \cdot \rho_w \cdot C_{pw}}{V_{\text{heated}} \cdot \rho C_{\text{composite}}} \tag{5}$$

3.2.2 Dynamic Viscosity Calculation

Viscosity $\mu(t)$ in Centipoise (cP) is computed continuously:

$$\mu(t) = \exp\left(A + \frac{B}{T(t) + 273.15}\right) \tag{6}$$

where constants A, B are calibrated via non-linear least squares fit on Baghewala laboratory PVT core reports for 18° API crude oil ($A = -14.28$, $B = 5820.4$).

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4 AI/ML Infrastructure & Diagnostic Pipeline

4.1 Computer Vision Dyno Card Diagnostic Engine

Downhole Dyno cards (F_{plunger} vs u_{plunger}) are converted into 224×224 single-channel grayscale raster tensors and processed by a lightweight convolutional network (ResNet-18) for automated anomaly classification.

Diagnostic State	Physical Signature on Downhole Card	Automated Control Response
Normal Operation	Full rectangular loop, clean valve opening/closing	Maintain target SPM trajectory
Rod Floating / Slack	Low force on downstroke top, compression loads	Trigger immediate SPM reduction by $\Delta\text{SPM} = -1.5$
Fluid Pound	Sharp force drop during downstroke prior to BDC	Reduce SPM / Decrease stroke rate to fill pump barrel
Gas Locking	Hyperbolic force decrease on downstroke, delayed valve	Cycle VFD speed to bleed gas; adjust soak timing
Pump Unsetting	Erratically shifted load baseline, severe load hysteresis	Emergency alert to field technicians; throttle SPM to safe min

4.2 Closed-Loop Control & Optimization Architecture (PPO Agent)

A Proximal Policy Optimization (PPO) Reinforcement Learning agent continuously calculates optimal VFD frequency commands $f_{\text{VFD}}(t)$.

- **State Space** S_t: $[T(t), \mu(t), \text{PPRL}_t, \text{RodStretch}_t, \text{FluidPoundRatio}_t, \text{CurrentSPM}_t, \text{SOR}_{\text{avg}}]$.
- **Action Space** A_t: VFD frequency adjust $\Delta f \in [-0.5\text{ Hz}, +0.5\text{ Hz}]$ bounded by safe operating limits $SPM_{\text{actual}} \leq SPM_{\text{safe}}(t)$.
- **Reward Function** R_t:

$$R_t = w_1 \cdot q_o(t) - w_2 \cdot P_{\text{elec}}(t) - w_3 \cdot \mathbb{I}(\text{RodFloat}) - w_4 \cdot |\sigma_{\text{peak}} - \sigma_{\text{allowable}}|^2 \tag{7}$$

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5 API Specifications & System Integration

5.1 gRPC Engine Interface Spec (digital_twin_engine.proto)

```
syntax = "proto3";
package baghewala.digitaltwin;

service TwinOptimizationService {
  rpc SolveWaveEquation (DynoCardRequest) returns (DynoCardResponse);
  rpc CalculateSafeSPM (ThermalStateRequest) returns (SafeSPMResponse);
  rpc EvaluateCycleEconomics (EconomicRequest) returns (EconomicResponse);
```

```
}

message DynoCardRequest {
  string well_id = 1;
  repeated double surface_position_in = 2;
  repeated double surface_load_lbs = 3;
  double fluid_viscosity_cp = 4;
  double rod_length_ft = 5;
}

message DynoCardResponse {
  repeated double plunger_position_in = 1;
  repeated double plunger_load_lbs = 2;
  bool rod_floating_detected = 3;
  double effective_stroke_length_in = 4;
  string diagnostic_classification = 5;
}

message ThermalStateRequest {
  string well_id = 1;
  double production_days = 2;
  double steam_injected_tons = 3;
}

message SafeSPMResponse {
  double max_safe_spm = 1;
  double recommended_vfd_hz = 2;
  double calculated_viscosity_cp = 3;
}
```

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6 Security, Compliance & Fail-Safe Engineering

6.1 Edge Fail-Safe Operating Modes

To protect subterranean equipment against communication loss or cloud service disruption, the wellsite Edge Gateway contains a fallback hardware watchdog:

1. **Heartbeat Supervision**: Edge gateway sends a 5-second ping to the cloud orchestration tier.
2. **Autonomous Degraded Mode**: If cloud connection drops for > 30 seconds, the edge gateway isolates control and executes a local simplified lookup table for $SPM_{\text{safe}}(T_{\text{wellhead}})$.
3. **Hard Mechanical Limit Overrides**: If surface load cell detects $PPRL > 90\%$ of rod tensile yield stress or compression load ≤ 0 lbs (extreme floating), the edge gateway triggers an immediate hard interrupt driving VFD frequency to minimum safe idle (2.0 SPM).

6.2 Security & Data Governance

- **Transport Security**: TLS 1.3 mutual authentication (mTLS) with X.509 certificates generated per edge device.
- **Role-Based Access Control (RBAC)**: Enforces write restrictions on VFD setpoint modifications (Only Level-3 Field Engineers and Autonomous Twin Core have actuation permissions).

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7 Infrastructure, Technology Stack & Deployment

7.1 Continuous Deployment (CI/CD) & Model Operations (MLOps)

- **Container Deployment**: Kubernetes (k8s) cluster managed with Helm charts, featuring automated HPA (Horizontal Pod Autoscaler) triggered at $> 70\%$ CPU utilization.

Subsystem Domain	Technology / Framework	Performance / Latency Specification
Edge Computing	Docker / C++20 / ONNX Runtime Edge	< 100 ms local inference & safe SPM check
Message Bus	Apache Kafka / EMQX Broker	$> 10,000$ msg/sec throughput, $<$ 10 ms latency
Numerical Solver	Python Numba / C++ Eigen Library	< 200 ms finite difference 1D wave solve
AI Model Serving	TorchScript / Triton Inference Server	< 150 ms Dyno card image classification
Time Series DB	TimescaleDB (PostgreSQL Extension)	$> 50,000$ inserts/sec, compressed storage
Frontend Visualization	React.js / Plotly.js WebGL rendering	60 FPS real-time Dyno card rendering

- **Model Retraining Pipeline**: Drift detection monitors trigger automated PyTorch model retraining when classification confidence drops below 88% over 50 consecutive pumping cycles.